

CDS505

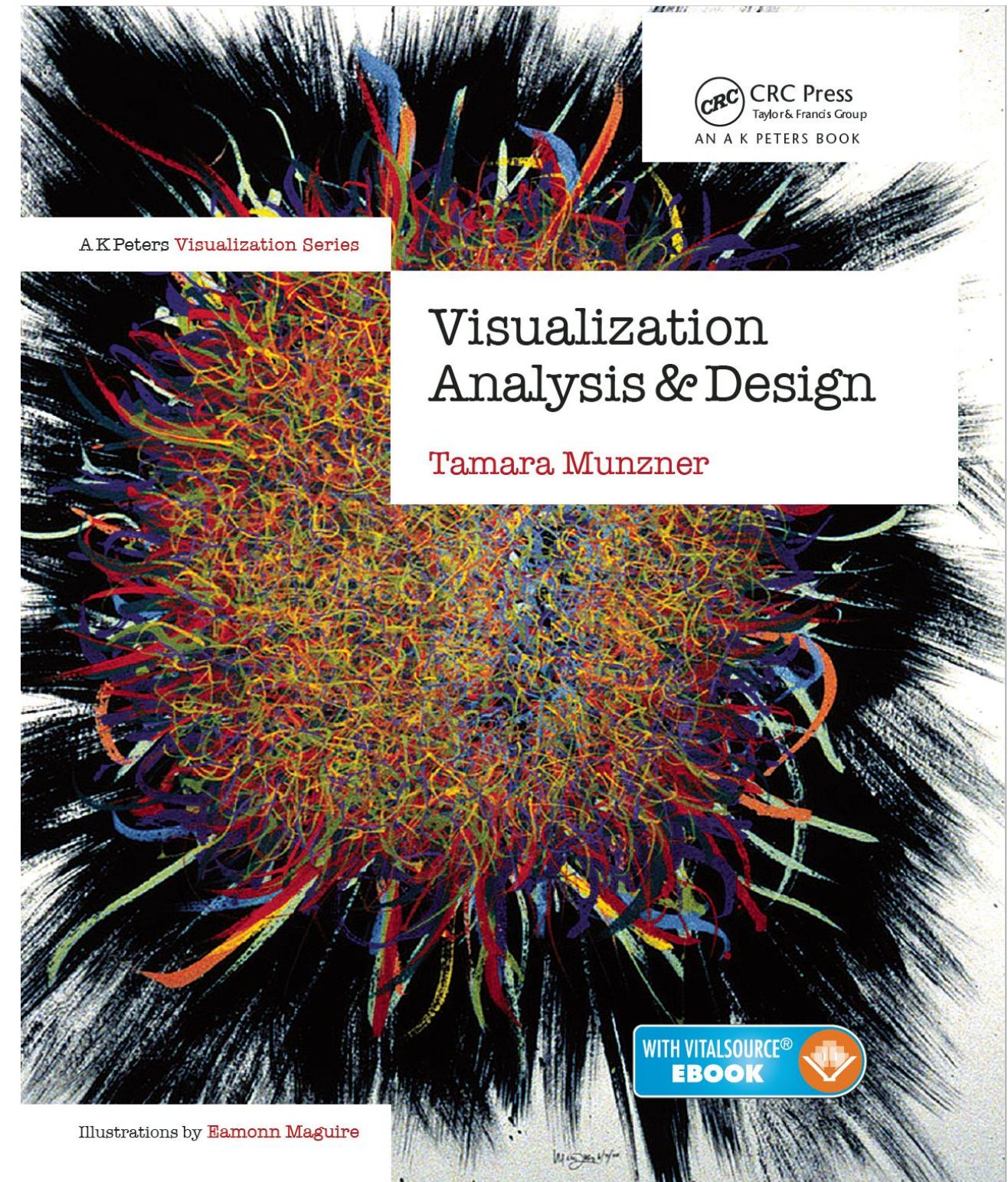
Data Visualization & Visual Analytics

Data Abstraction

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Universiti Sains Malaysia*

Website:

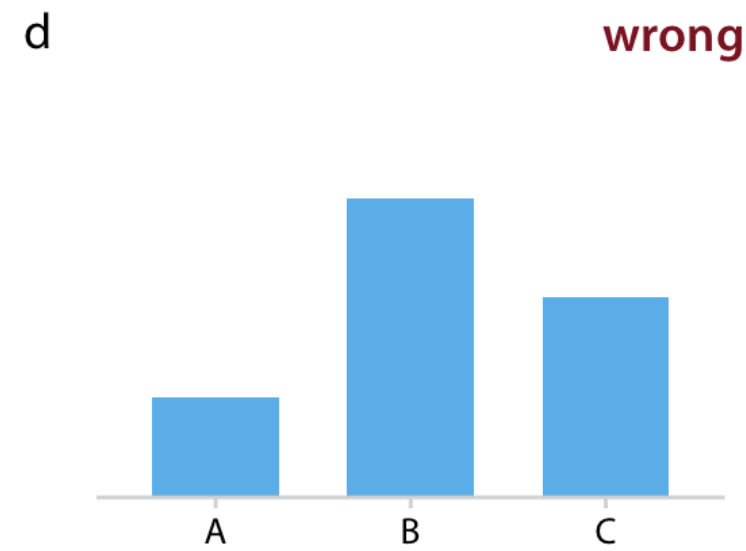
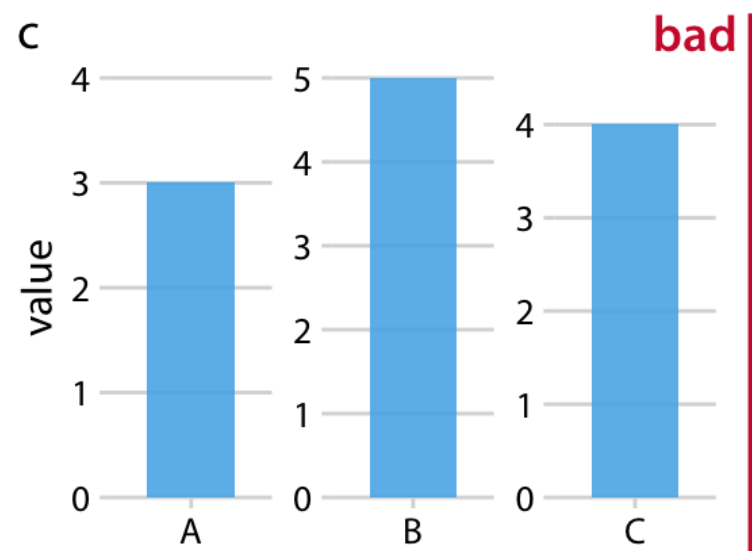
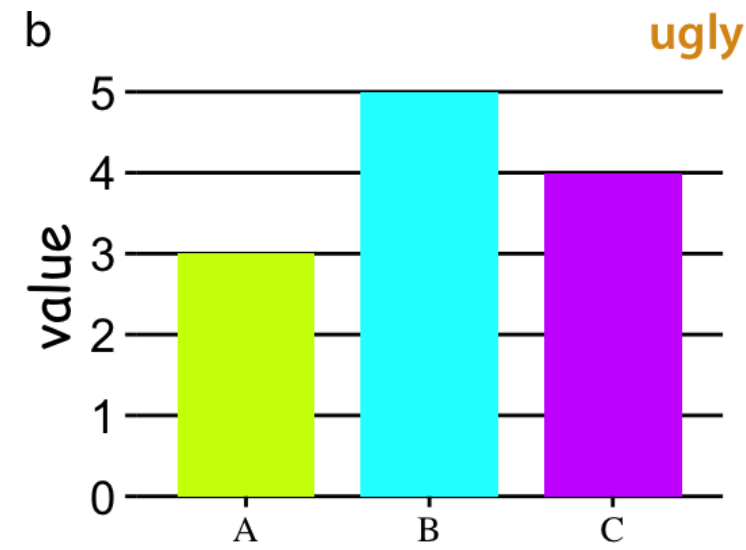
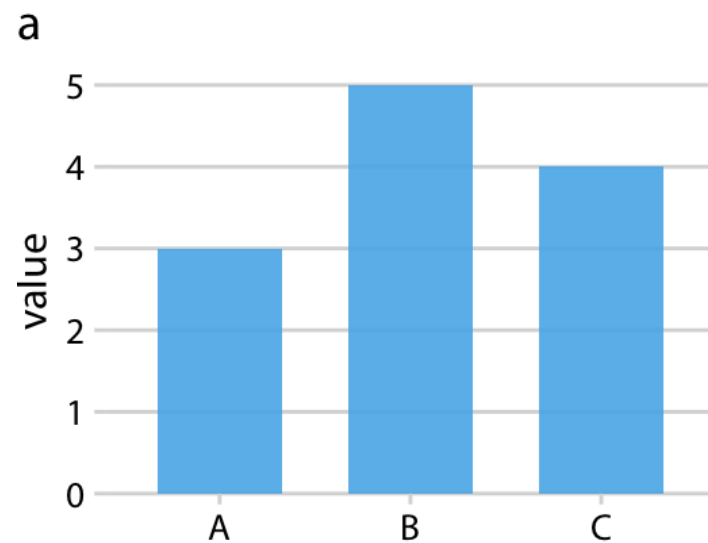
<https://cs.usm.my/index.php/faculty-member/194-nur-intan-raihana-ruhaiyem-dr>



- ✓ Recap previous class
- ✓ Data Abstraction
- ✓ Items and Attributes
- ✓ Class Activity

Class Activity: Discussion

RECAP



- a) A bar plot showing three values ($A = 3$, $B = 5$, and $C = 4$). This is a reasonable visualization with no major flaws.
- b) An **ugly** version of part (a). While the plot is technically correct, it is not aesthetically pleasing. The colors are too bright and not useful. The background grid is too prominent. The text - using three different fonts in three different sizes.
- c) A **bad** version of part (a). Each bar is shown with its own y-axis scale. Because the scales don't align, this makes the figure misleading.
- d) A **wrong** version of part (a). Without an explicit y axis scale, the numbers represented by the bars cannot be ascertained. The bars appear to be of lengths 1, 3, and 2, even though the values displayed are meant to be 3, 5, and 4.

RECAP

Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively.

Why?...

Why represent all the data?

RECAP

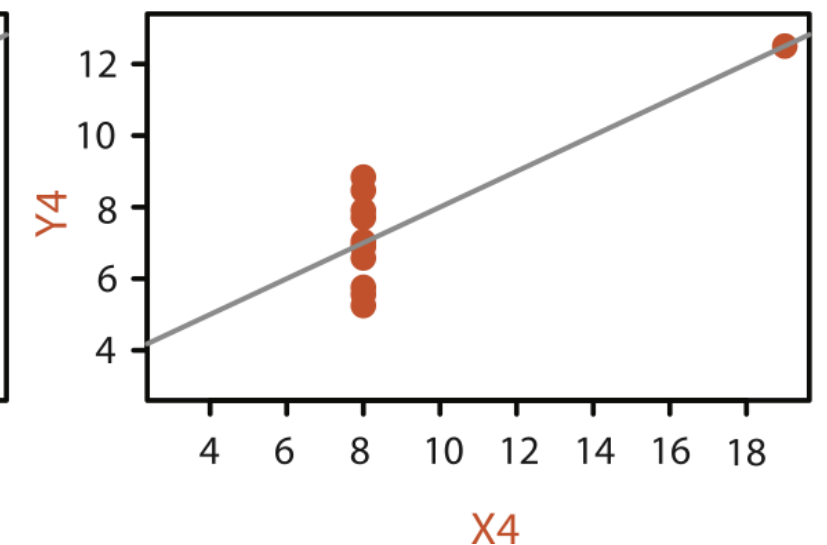
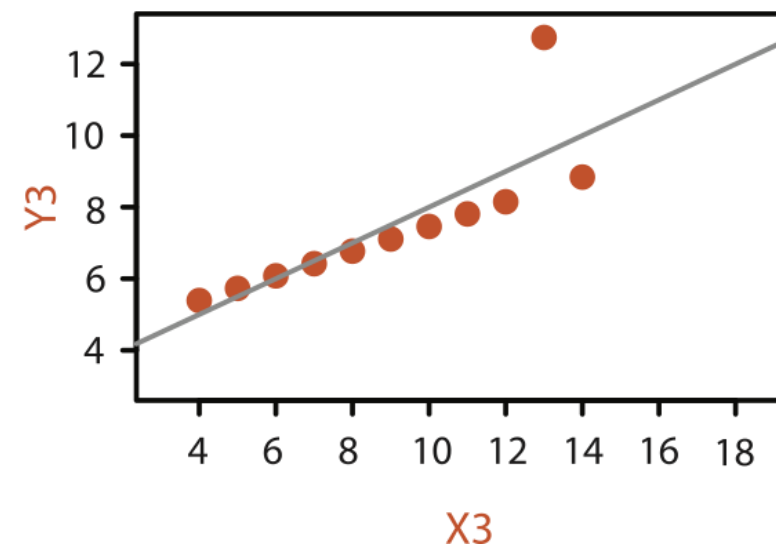
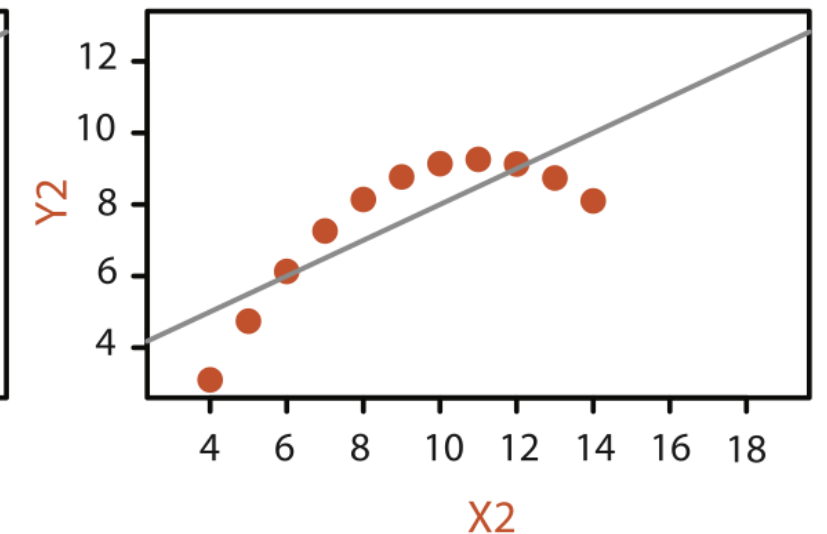
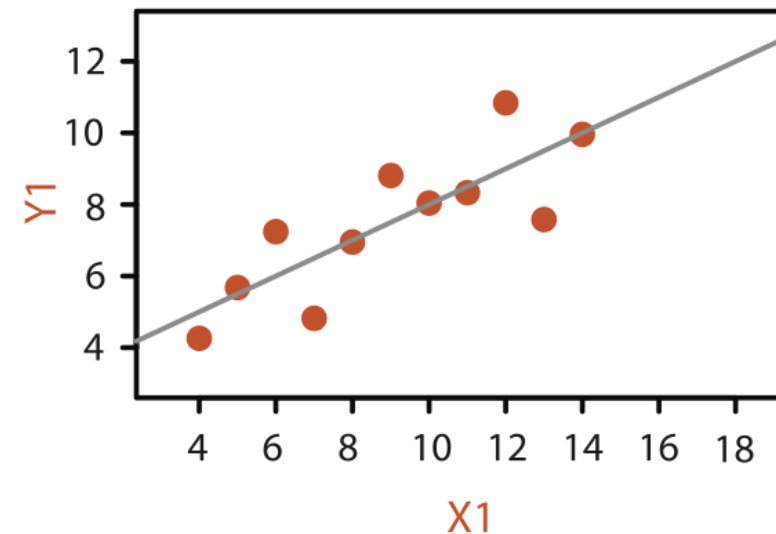
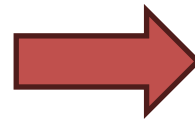
Summaries lose information, details matter

- confirm expected and find unexpected patterns
- assess validity of statistical model

Anscombe's Quartet

Identical statistics

x mean	9
x variance	10
y mean	7.5
y variance	3.75
x/y correlation	0.816



What resource limitations are we faced with?

RECAP

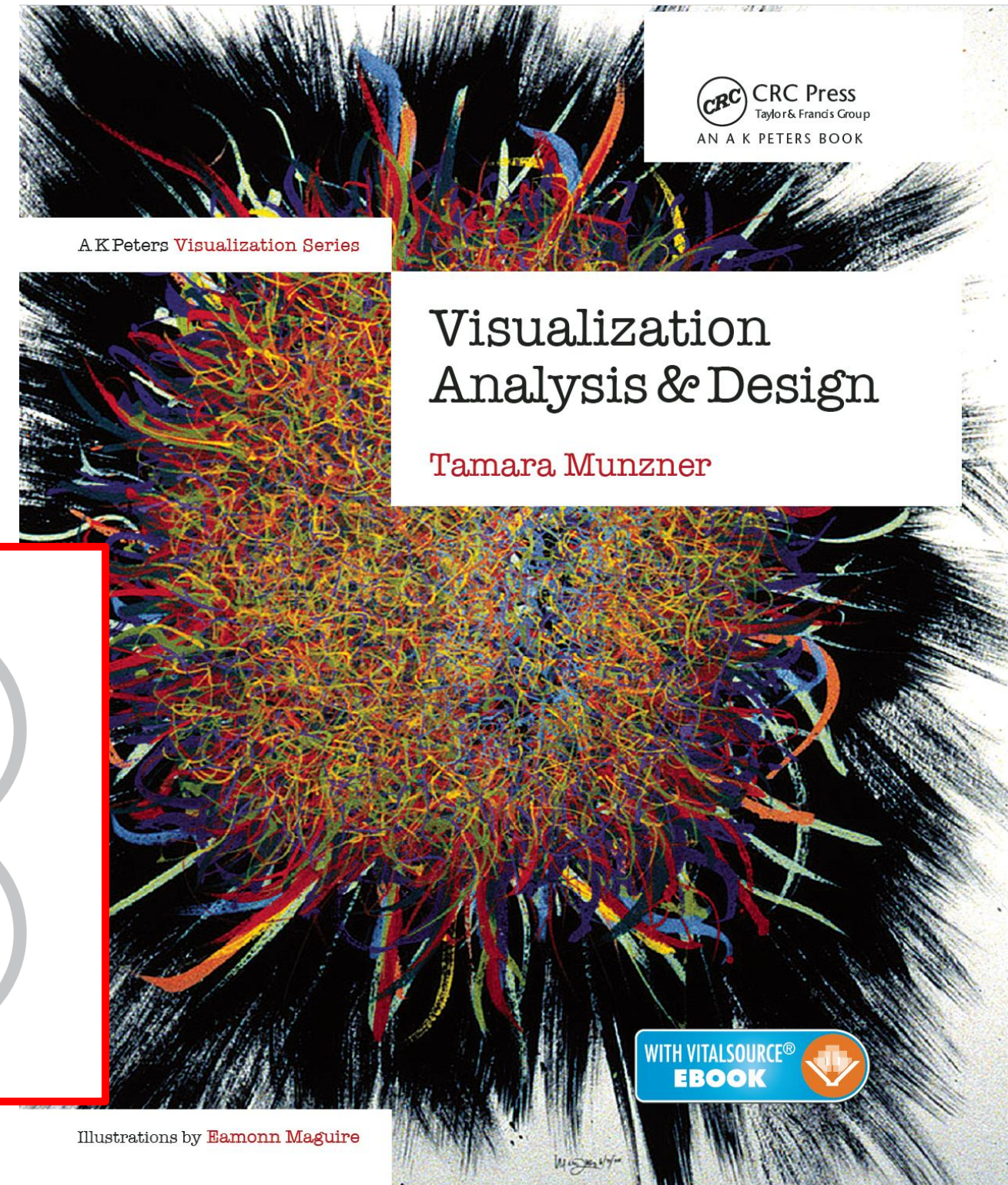
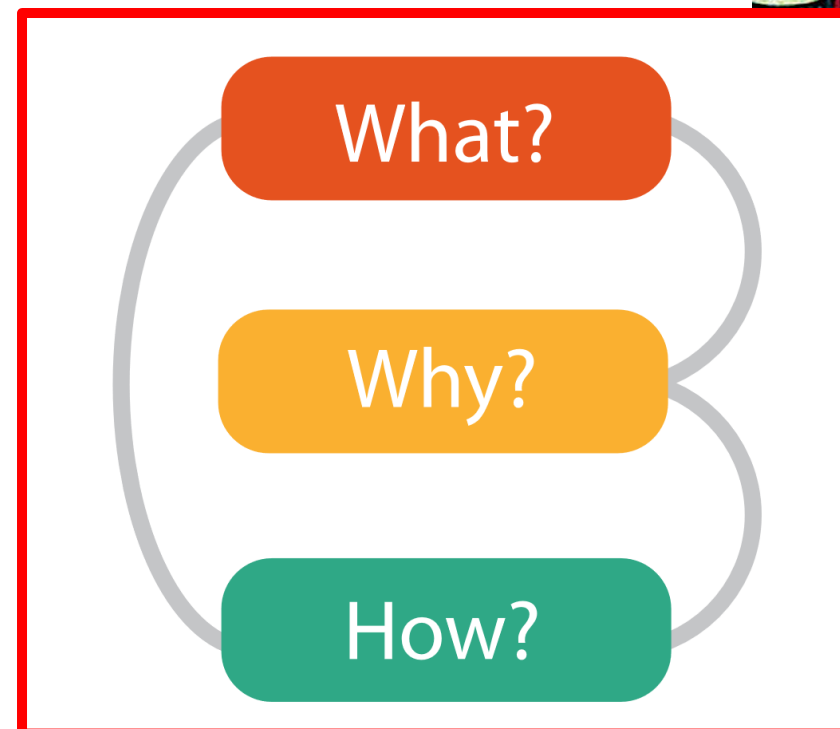
Vis designers must take into account three very different kinds of resource limitations: those of computers, of humans, and of displays.

- computational limits
 - computation time, system memory
- display limits
 - pixels are precious & most constrained resource
 - **information density**: ratio of space used to encode info vs unused whitespace
 - tradeoff between clutter and wasting space
 - find sweet spot between dense and sparse
- human limits
 - human time, human memory, human attention

Visualisation Analysis & Design (Tamara Munzner)

RECAP

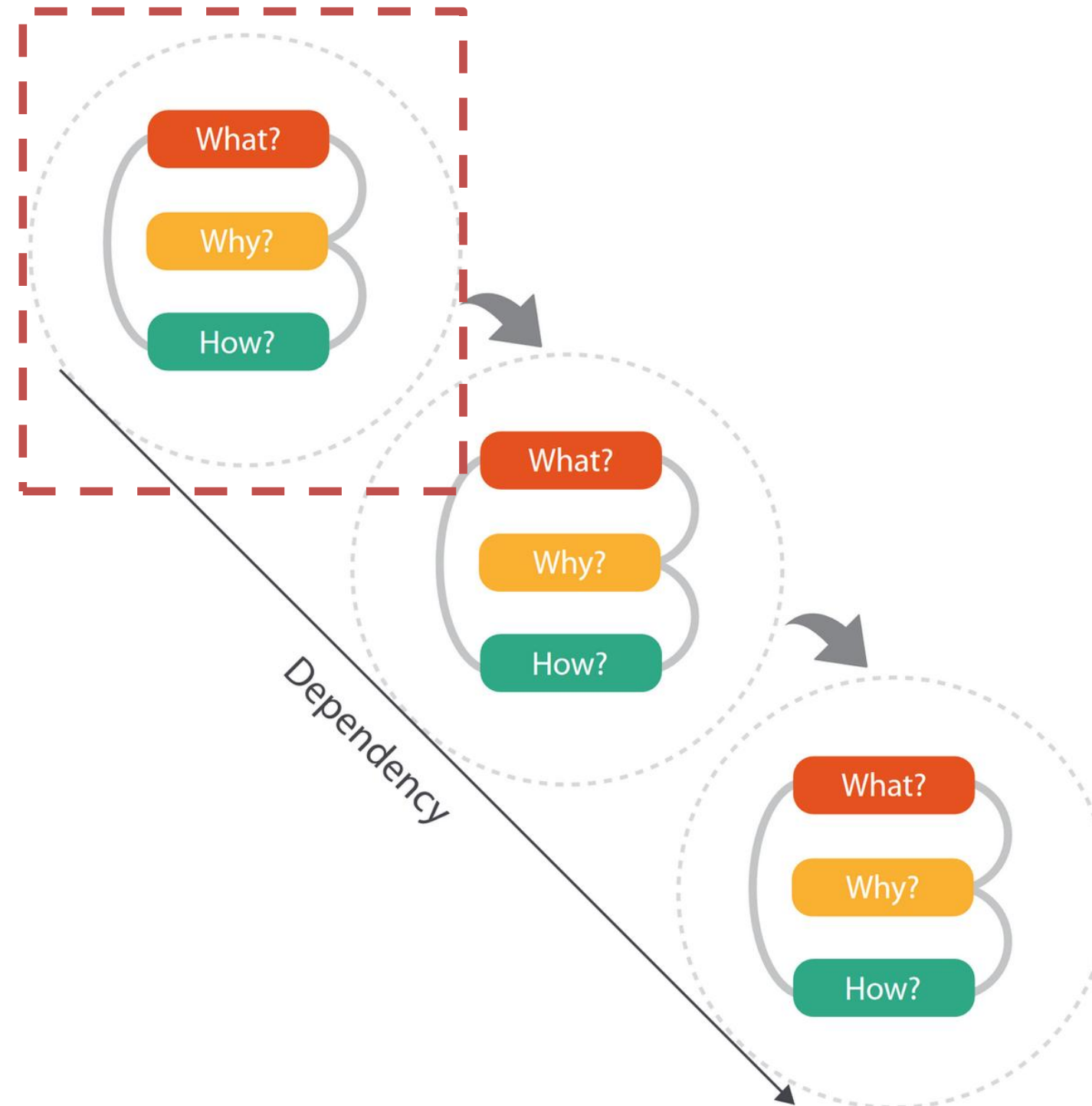
- Theoretical framework – helps understand why standard answer is ‘it depends’
- Deeper into the concepts, the jargon and research of Data Vis
- Very well structured



The instances of the What-Why-How Framework

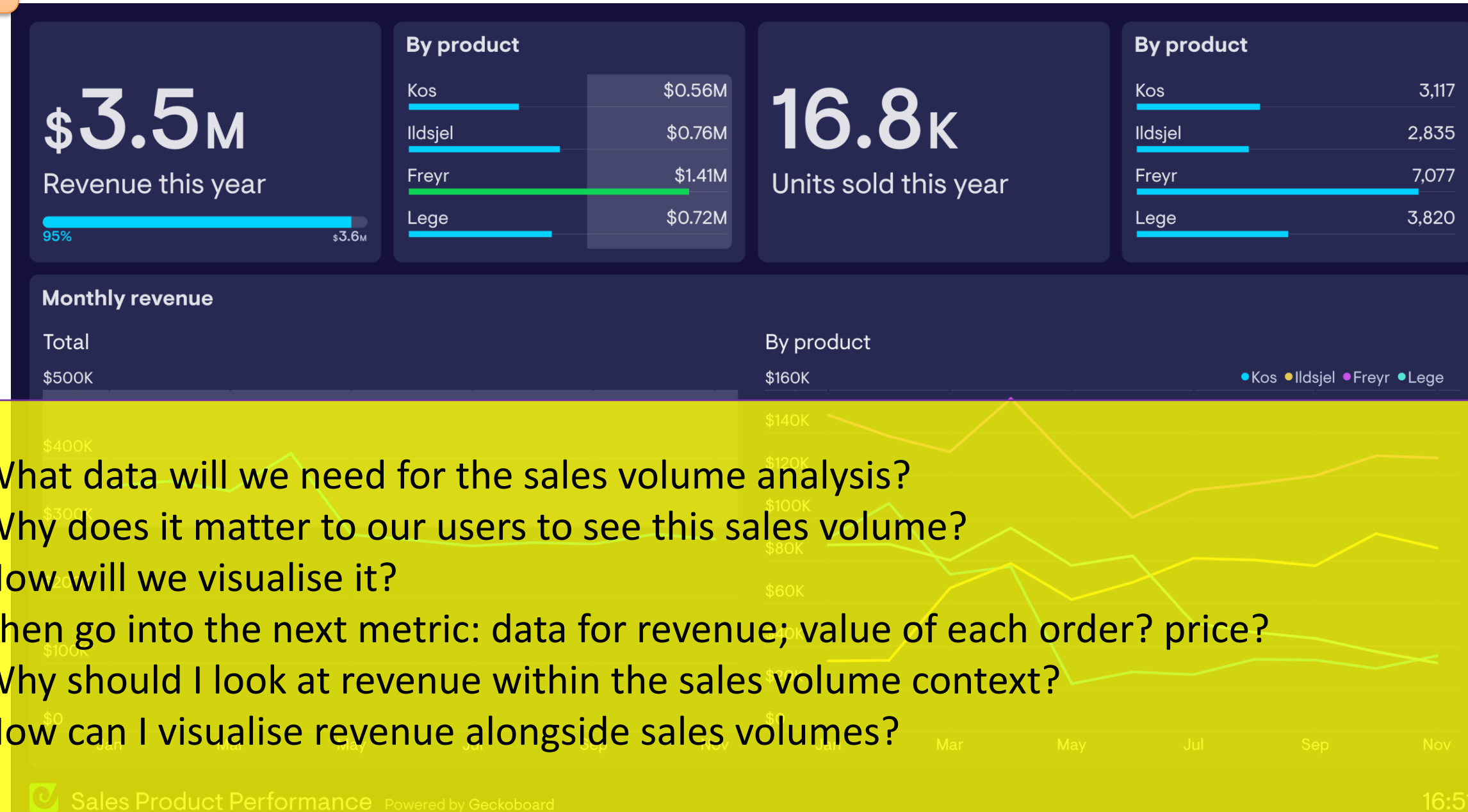
RECAP

One instance



Example of Sales Dashboard

RECAP



- What data will we need for the sales volume analysis?
- Why does it matter to our users to see this sales volume?
- How will we visualise it?
- Then go into the next metric: data for revenue; value of each order? price?
- Why should I look at revenue within the sales volume context?
- How can I visualise revenue alongside sales volumes?

We then go through the same basic framework for profit and any further visualisations or themes we should add.

RECAP

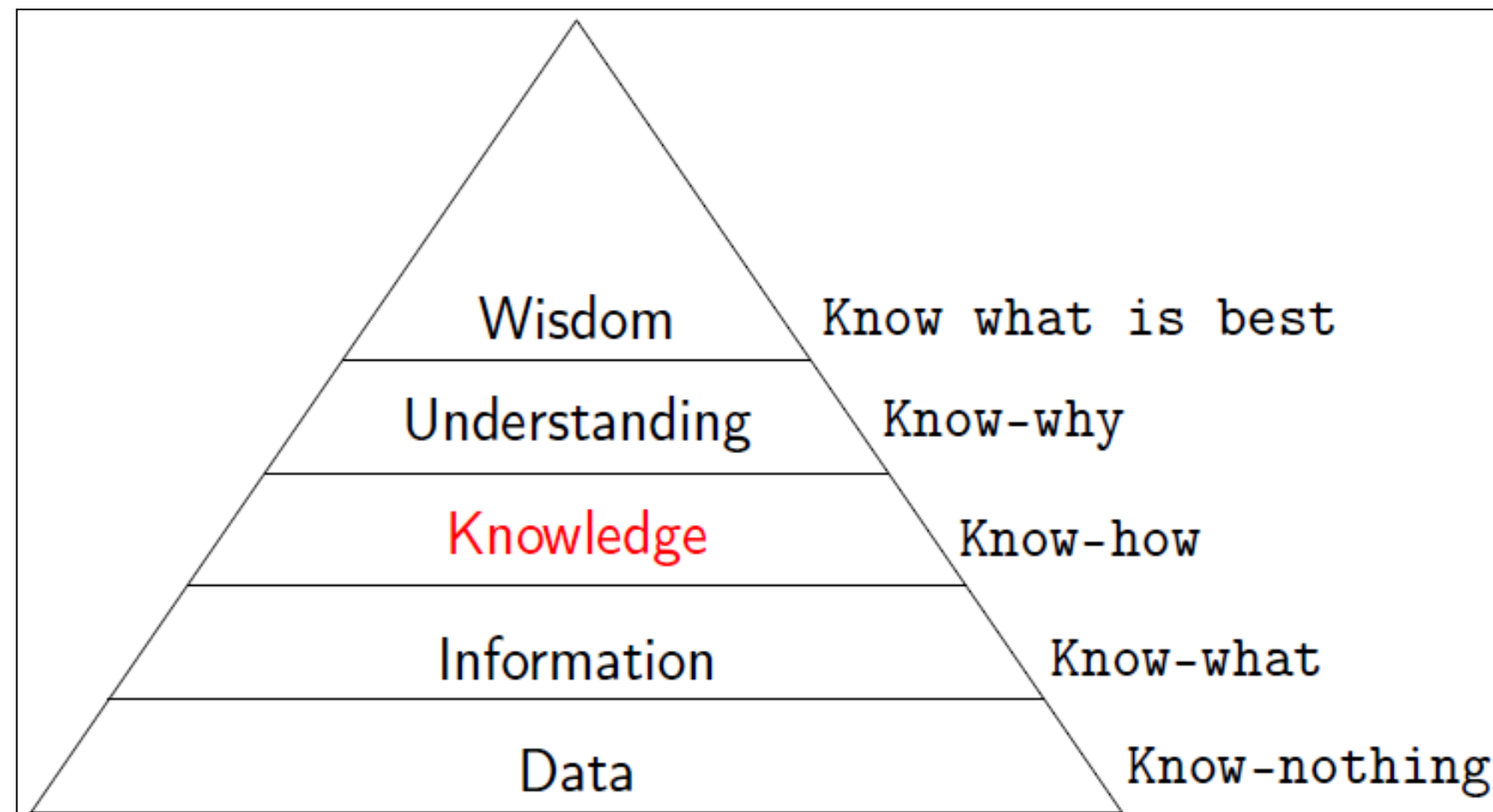
1. Find **two different visualizations** (on the internet, in newspapers, TV, etc.). Different means, e.g., a line chart, a node-link diagram.
2. For each visualization, **describe three faults** and **why** they are **problematic**. Make sure to report different faults for each visualization, i.e., in **total you should report six faults**.
3. For each visualization, propose (sketch) one or two visualizations that solve the identified faults.
4. For each solution you find, argue with a more **general** visualization **guidelines**.

Homework Discussion

Data Abstraction: What can be visualised?

What does data mean?

- What is your understanding about data?
- Think about what are the common terms used to describe data.



What does data mean?

- What is your understanding about data?
- Think about what are the common terms used to describe data
- Data in itself is meaningless or ambiguous

14, 2.6, 30, 30, 15, 100001

What does this sequence of six numbers mean?

- two points far from each other in 3D space?
- two points close to each other in 2D space, with 15 links between them, and a weight of 100001 for the link?
- something else??

Basil, 7, S, Pear

What about this data?

- food shipment of produce (basil & pear) arrived in satisfactory condition on 7th day of month
 - Basil Point neighborhood of city had 7 inches of snow cleared by the Pear Creek Limited snow removal service
 - lab rat Basil made 7 attempts to find way through south section of maze, these trials used pear as reward food
- Data is what we want to visualize (input to our viz tool)
 - Data influence many aspect of our viz design

Now what?

- Semantic: real world meaning
- Data types: structural or mathematical interpretation of data
 - item, link, attribute, position, (grid)
 - different from data types in programming!

Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

Tables

Items

Attributes

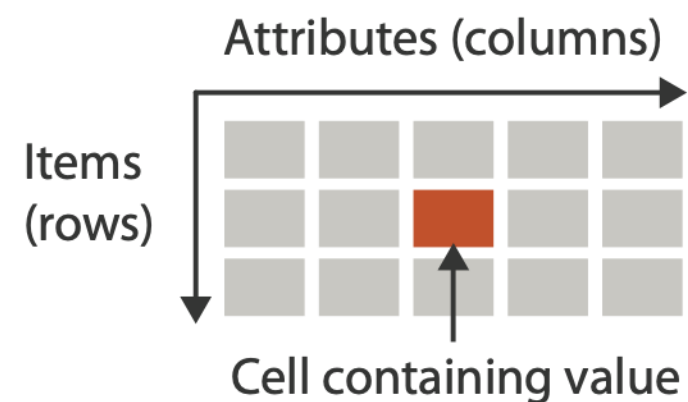
- **Flat table**
 - one item per row
 - each column is attribute
 - cell holds value for item-attribute pair
 - unique key (could be implicit)

attributes: name, age, shirt size, fav fruit

ID	Name	Age	Shirt Size	Favorite Fruit
1	Amy	8	S	Apple
2	Basil	7	S	Pear
3	Clara	9	M	Durian
4	Desmond	13	L	Elderberry
5	Ernest	12	L	Peach
6	Fanny	10	S	Lychee
7	George	9	M	Orange
8	Hector	8	L	Loquat
9	Ida	10	M	Pear
10	Amy	12	M	Orange

item: person

→ Tables



attributes: name, age, shirt size, fav fruit

- **Item:** individual entity, discrete
 - e.g., patient, car, stock, city
 - "independent variable"
- **Attribute:** property that is measured, observed, logged...
 - e.g., height, blood pressure for patient
 - e.g., horsepower, make for car
 - "dependent variable"

Name	Age	Shirt Size	Favorite Fruit
Amy	8	S	Apple
Basil	7	S	Pear
Clara	9	M	Durian
Desmond	13	L	Elderberry
Ernest	12	L	Peach
Fanny	10	S	Lychee
George	9	M	Orange
Hector	8	L	Loquat
Ida	10	M	Pear
Amy	12	M	Orange

item: person

- **Links**
 - express relationship between two items
 - e.g. friendship on facebook, interaction between proteins
- **Positions**
 - spatial data: location in 2D or 3D
 - pixels in photo, voxels in MRI scan, latitude/longitude
- **Grids**
 - sampling strategy for continuous data

Table

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.5	2/22/08
32	7/16/07	2-High	Small Pack	0.72	7/17/07
32	7/16/07	2-High	Jumbo Box	0.6	7/17/07
32	7/16/07	2-High	Medium Box	0.65	7/18/07
32	7/16/07	2-High	Medium Box	0.52	7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.58	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.55	10/25/07
36	11/3/07	1-Urgent	Small Box	0.49	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.56	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.44	1/20/05
69	6/4/05	4-Not Specified	Small Pack	0.6	6/6/05
69	6/4/05	4-Not Specified	Wrap Bag	0.59	6/6/05
70	12/18/06	5-Low	Small Box	0.82	12/23/06
70	12/18/06	5-Low	Wrap Bag	0.55	12/23/06
96	4/17/05	2-High	Small Box	0.38	4/19/05
97	1/29/06	3-Medium	Small Box	0.37	1/30/06
129	11/19/08	5-Low	Small Box	0.37	11/28/08
130	5/8/08	2-High	Small Box	0.38	5/9/08
130	5/8/08	2-High	Medium Box	0.6	5/10/08
130	5/8/08	2-High	Small Box	0.6	5/11/08
132	6/11/06	3-Medium	Medium Box	0.69	6/12/06
132	6/11/06	3-Medium	Jumbo Box	0.82	6/14/06
134	5/1/08	4-Not Specified	Large Box	0.64	5/3/08
135	10/21/07	4-Not Specified	Small Pack	0.55	10/23/07
166	9/12/07	2-High	Small Box	0.57	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.42	8/10/06
194	4/5/08	3-Medium	Wrap Bag		4/7/08

attribute

cell

item

Dataset Types

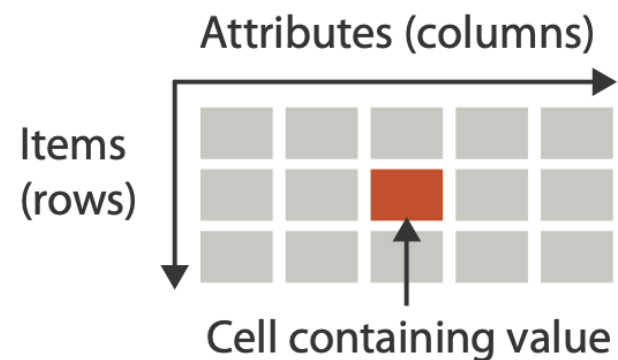
Tables

Items

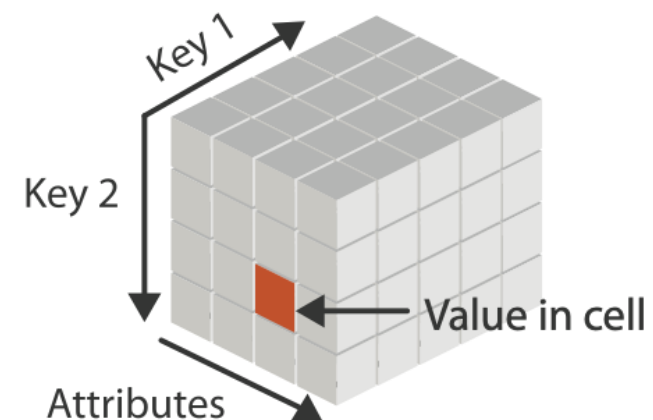
Attributes

- **Multidimensional tables**
 - indexing based on multiple keys
 - E.g. genes, patients

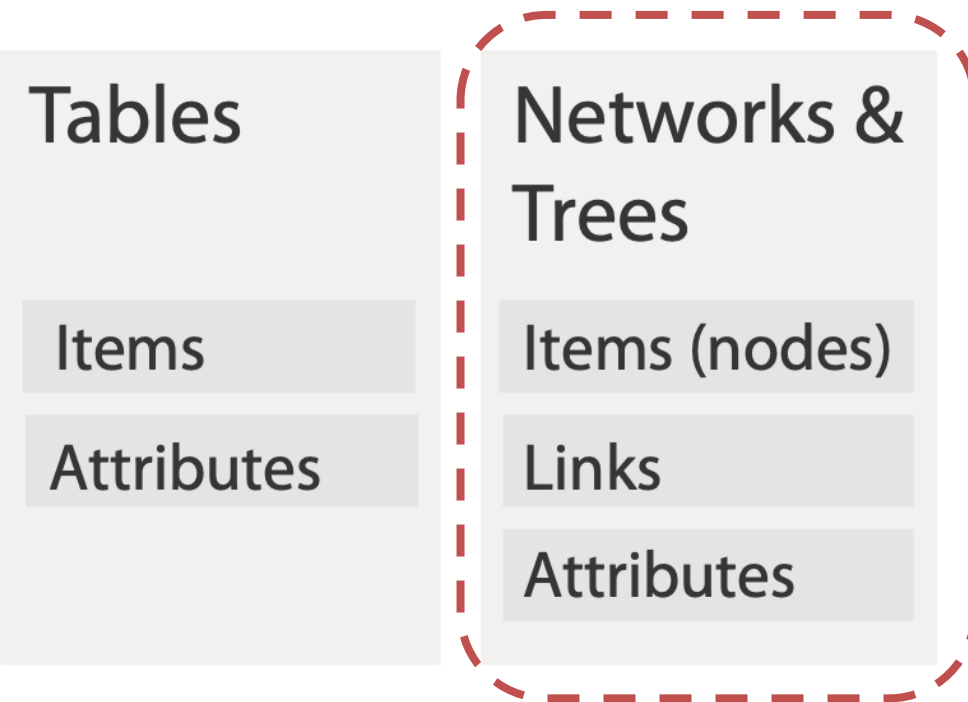
→ Tables



→ *Multidimensional Table*



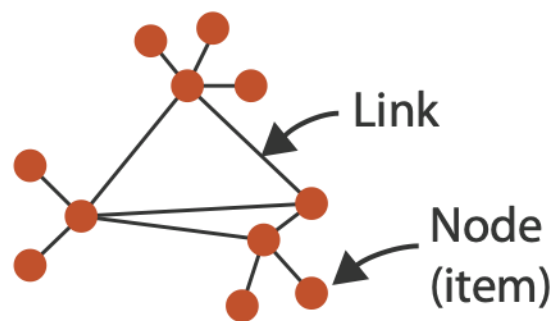
	A	B	C	D	E
1	#	A	B	C	D
2	1	#	A	B	C
3	2	1	#	A	B
4	3	2	1	#	A
5	4	3	G	2	#1.2
6	5	4	L	3	GeneName
7	6	5	P	4	LTF
8	7	6	T	5	POSTN
9	8	7	H	6	TMSL8
10	9	8	R	7	HLA-DQA1
11	10	9	S	8	RP11-35N6.1
12	11	10	D	9	STMN2
13	12	11	A	10	DCX
14	13	12	IL	11	AGXT2L1
15	14	13	SI	12	IL13RA2
16	15	14	M	13	SLN
17	16	15	C	14	MEOX2
18	17	16	N	15	COL11A1
19	18	17	F	16	NNMT
20	19	18	C	17	F13A1
21	20	19	M	18	CXCL14
22	21	20	T	19	MBP
	22	21	K	20	TF
		22	G	21	KCND2



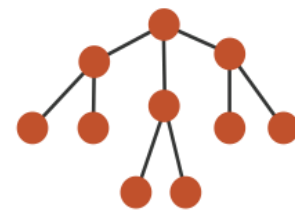
- **Network/graph**

- nodes (vertices) connected by links (edges)
- tree is special case: no cycles
 - often have roots and are directed

→ Networks



→ Trees



- Suitable for specifying or describing **relationship** between items (nodes)
- E.g. Social network : node a person, link contacts or friends of a person

Dataset Types

Tables

Items

Attributes

Networks & Trees

Items (nodes)

Links

Attributes

Fields

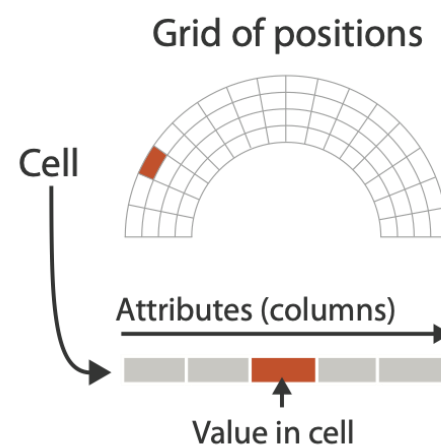
Grids

Positions

Attributes

→ Spatial

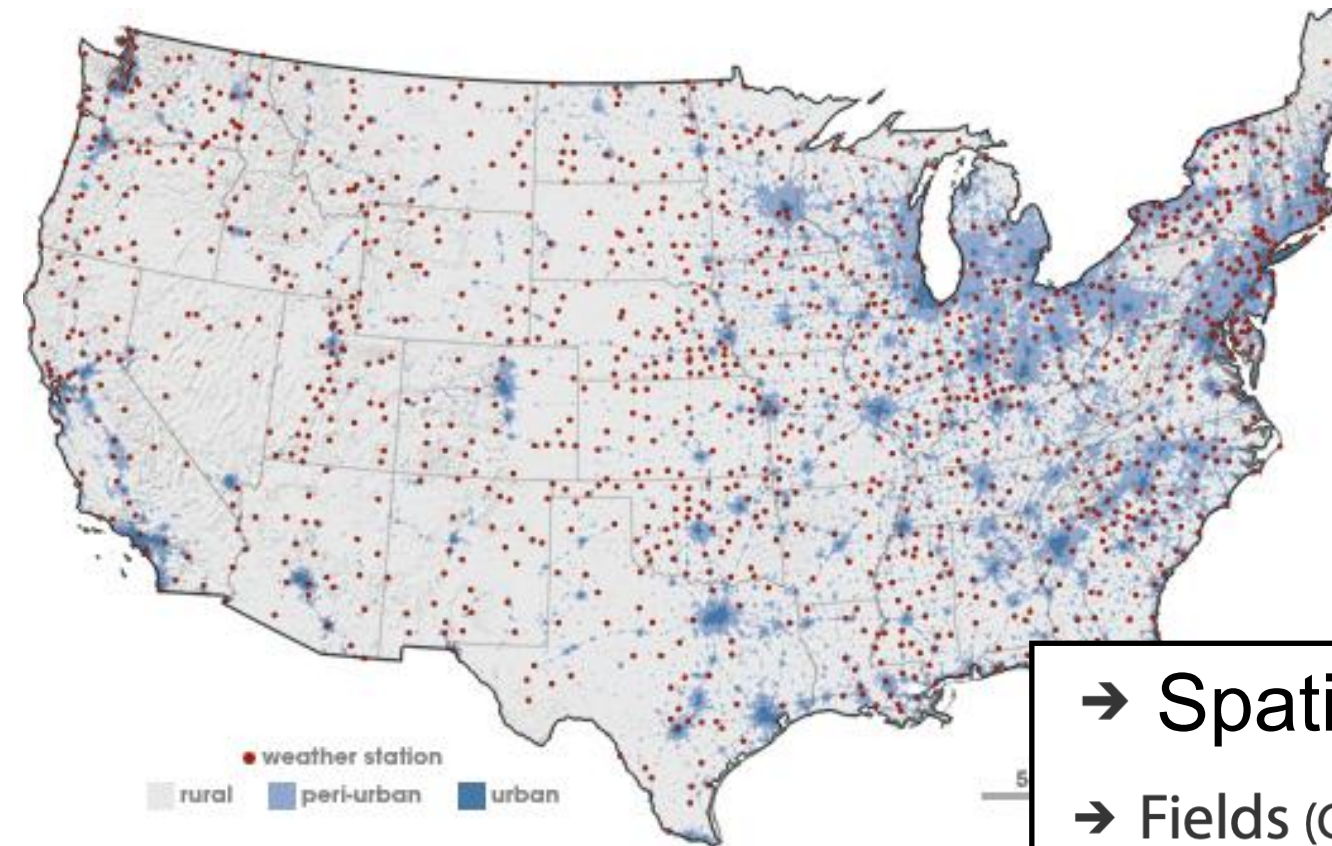
→ Fields (Continuous)



- Suitable for specifying *continuous* fields i.e. recording physical phenomena
- E.g. temperature over an airplane's wing, a curvilinear grid (2D) over which temperature data are measured/simulated

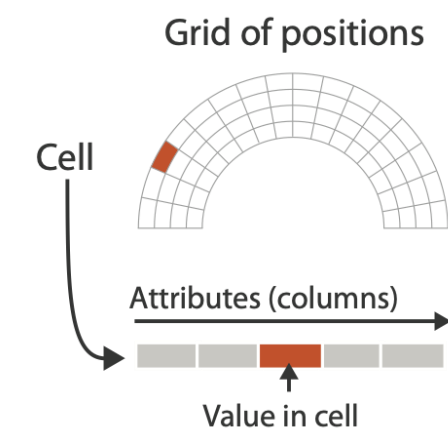
Spatial Fields

- Attribute values associated w/ cells
- Cell contains value from continuous domain
 - E.g. temperature, pressure, wind velocity
- Measured or simulated

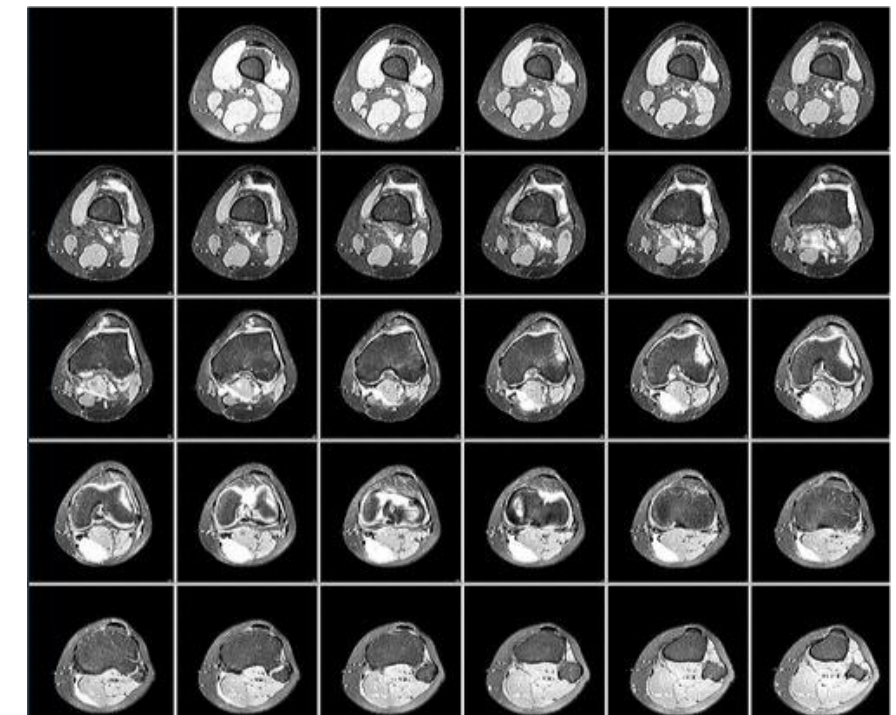
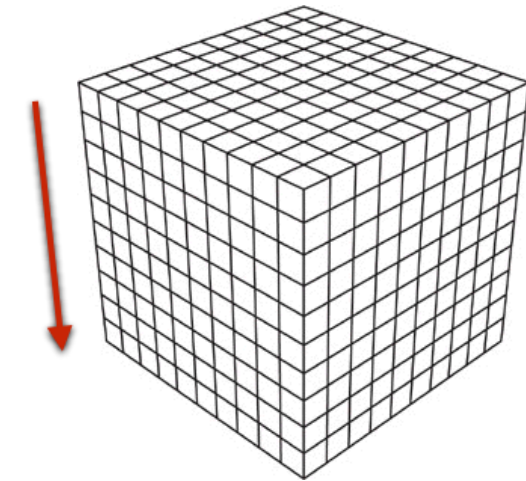


→ Spatial

→ Fields (Continuous)



- Attribute values associated w/ cells
- Cell contains value from continuous domain
 - eg temperature, pressure, wind velocity
- Measured or simulated
- Major concerns
 - **sampling:**
where attributes are measured
 - **interpolation:**
how to model attributes elsewhere
 - **grid types**

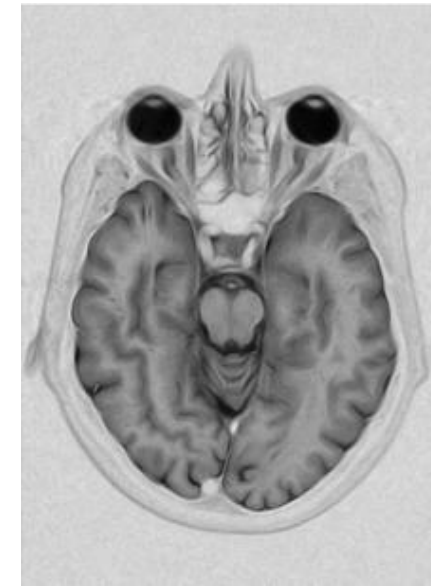


- Major divisions
 - attributes per cell:
scalar (1), vector (2), tensor (many)

Scalar is univariate, a single value attribute at a point in space. Examples : a temperature at a 3D location in a room.

Vector is multivariate, a list of multiple attribute values at each point in space. Examples : the velocity of air on the surface of earth (3D vector). Two quantities : magnitude & direction

Tensor is multivariate, also multiple attribute values at each point in space (generally $> 3D$ vectors). Examples : neuron fiber crossing in the white-matter of the brain



scalar

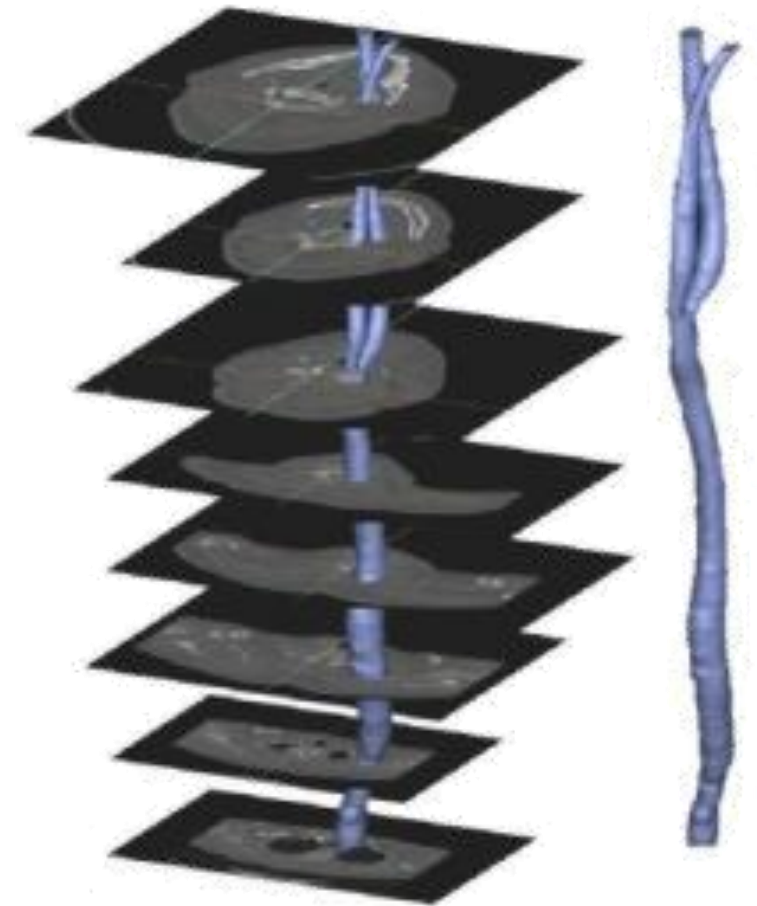
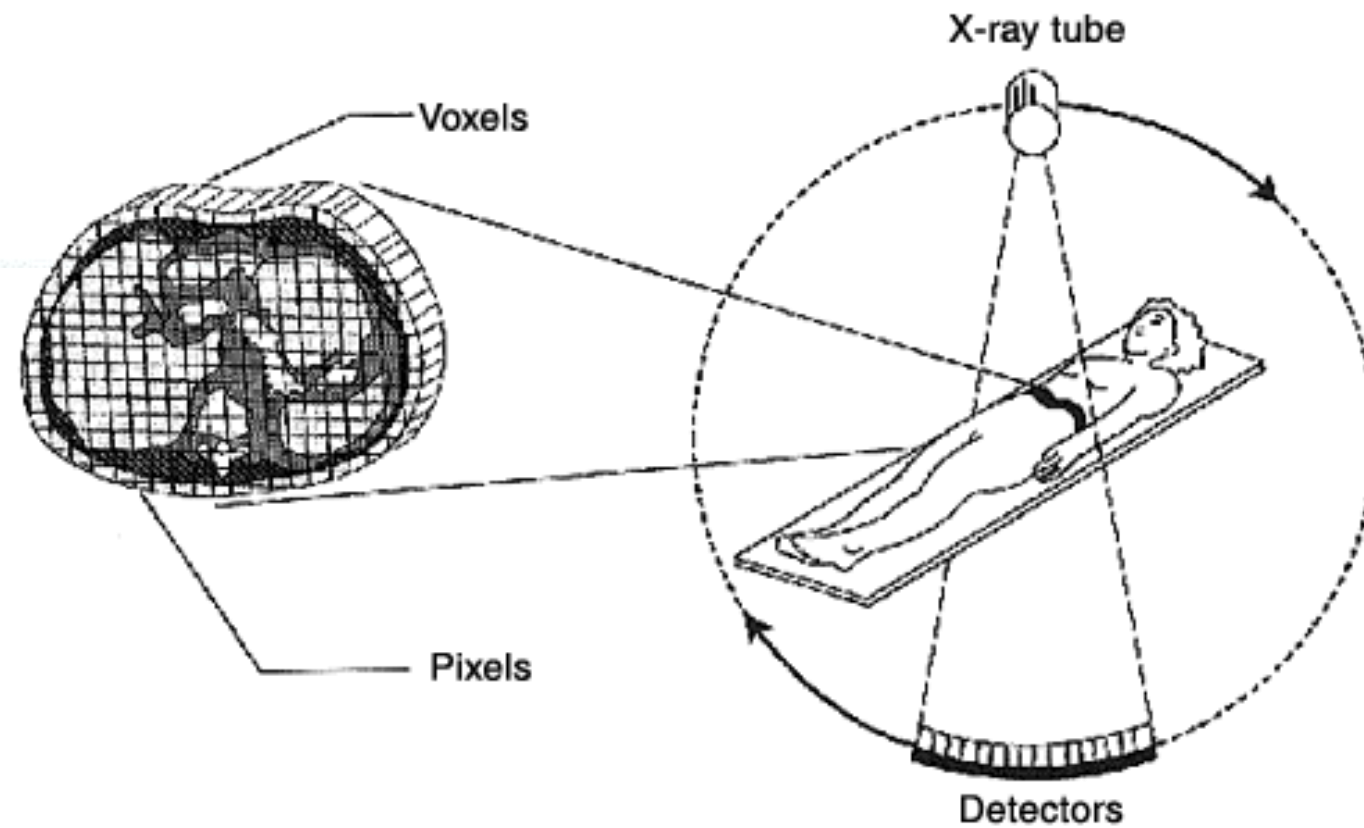
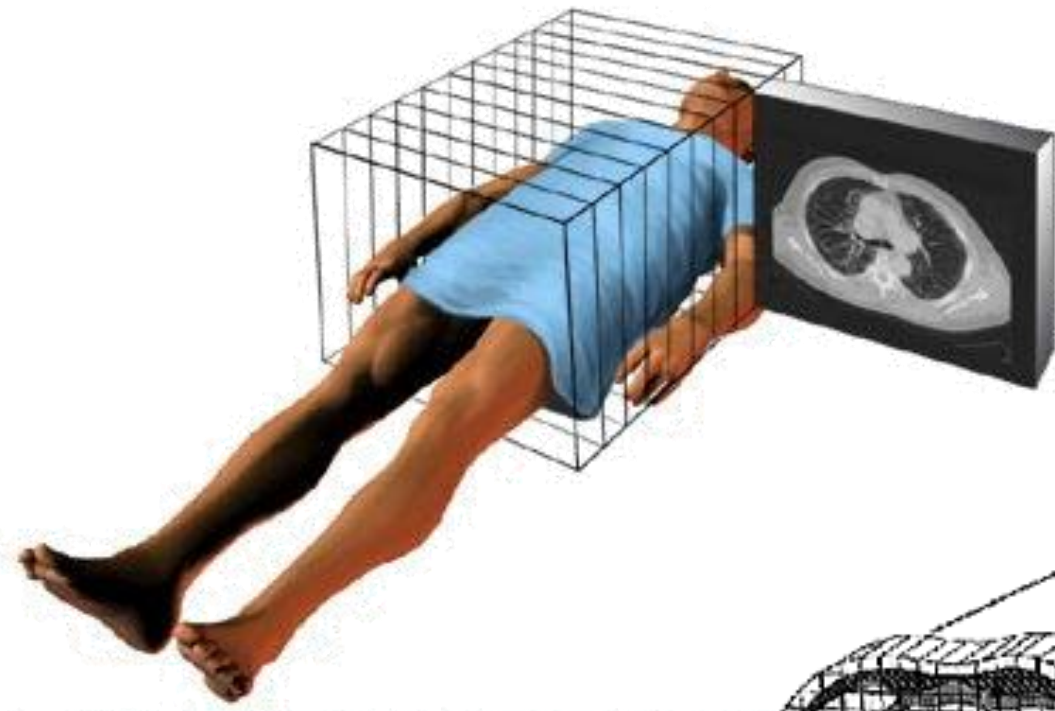


vector



tensor

Example of Fields Dataset: Medical



Dataset Types

Tables

Items

Attributes

Networks & Trees

Items (nodes)

Links

Attributes

Fields

Grids

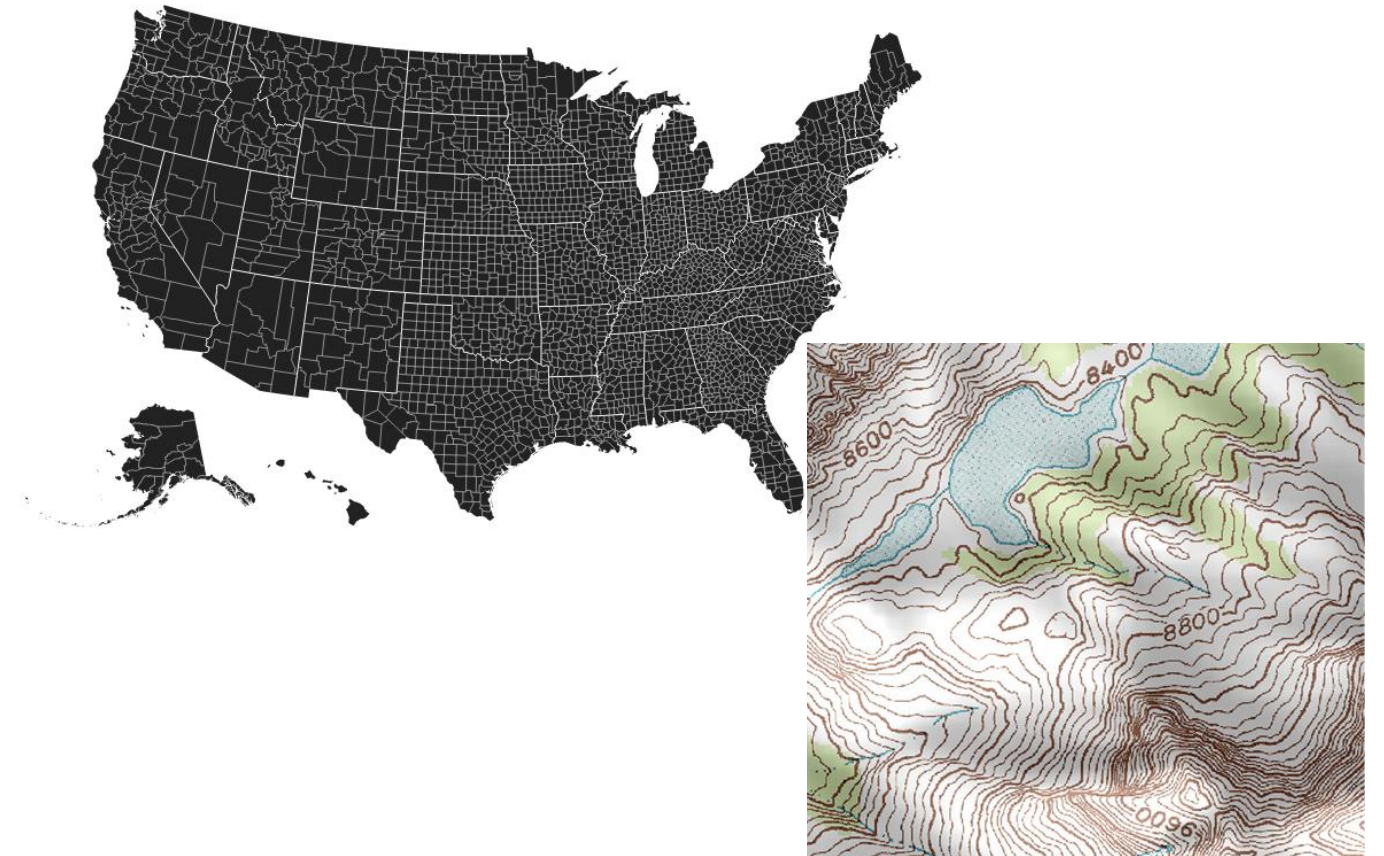
Positions

Attributes

Geometry

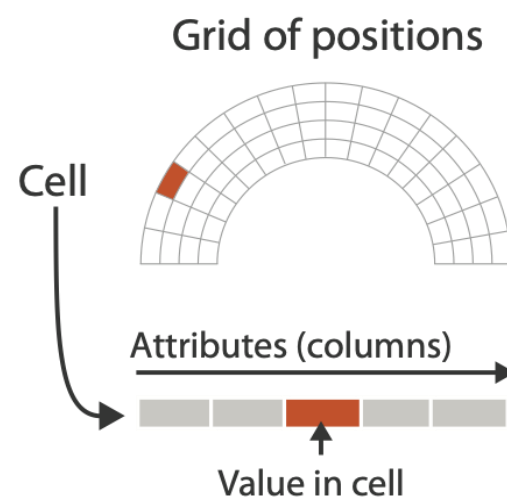
Items

Positions



→ Spatial

→ Fields (Continuous)



→ Geometry (Spatial)



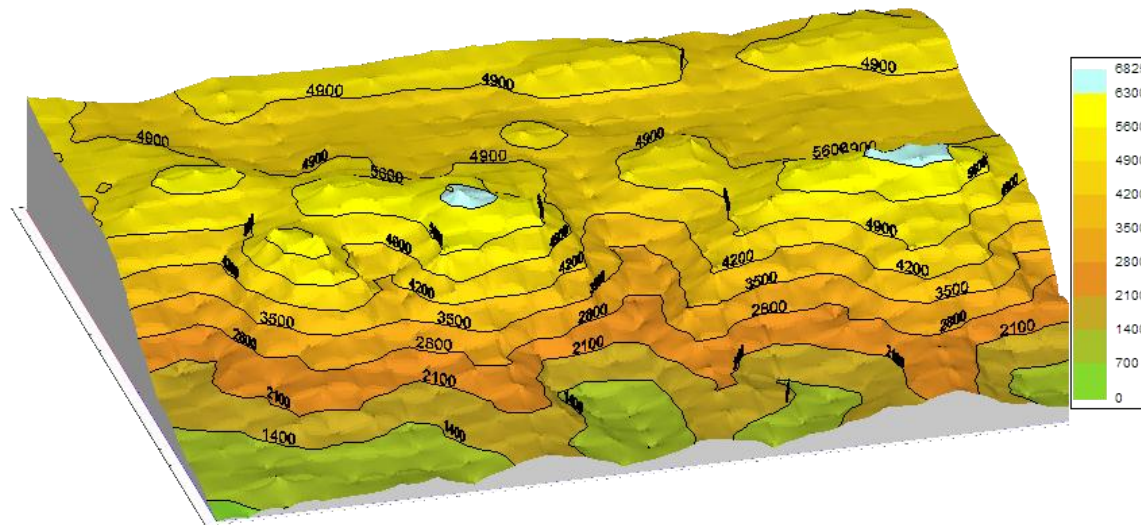
- Geometry is computer graphics core area e.g. point, line, polygon, colour, texture, etc.
- CG is use to synthesize geometry objects
- E.g. visual cues about the data overlaid (red dots – geometry objects themselves) over the map

- Shape of items
- Explicit spatial positions / regions
 - points, lines, curves, surfaces, volumes
- Boundary between computer graphics and visualization
 - **graphics**: geometry taken as given
 - **vis**: geometry is result of a design decision

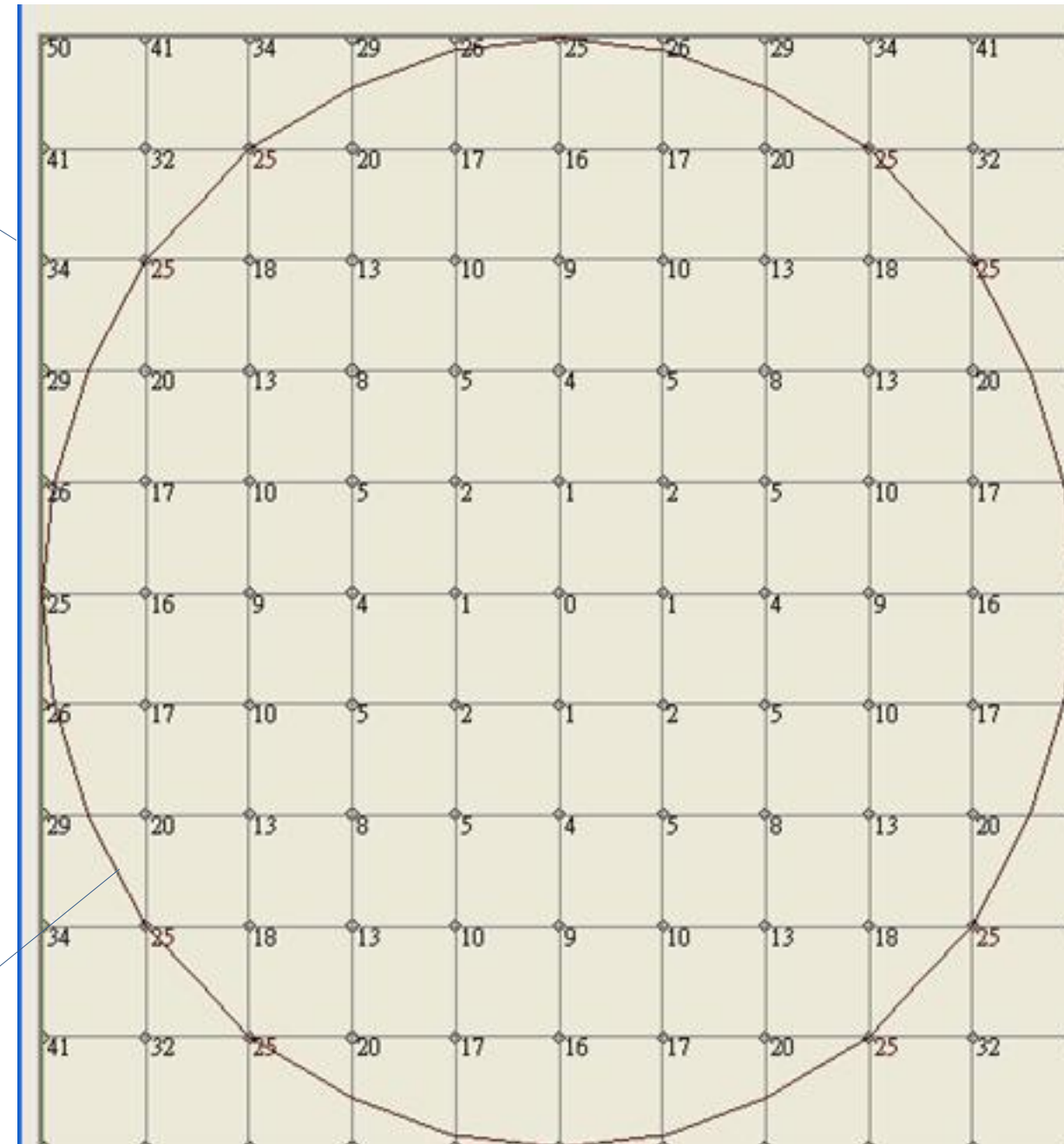


Geometry Dataset example: Contour

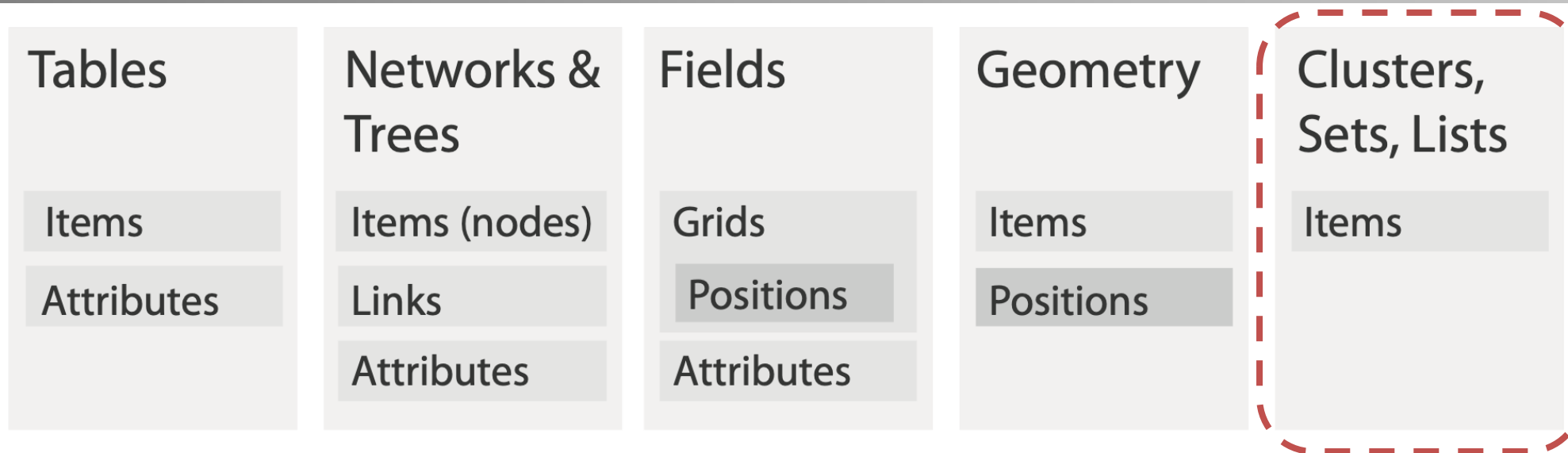
Spatial field dataset :
uniform grid, height
(attribute)



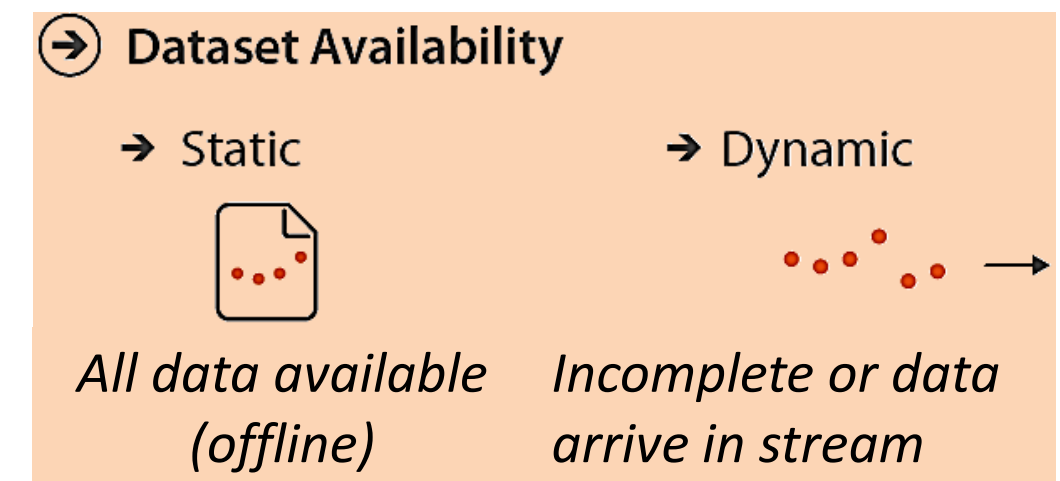
Geometry dataset:
Isoline encodes height =
25 extracted from field
dataset



Dataset Types



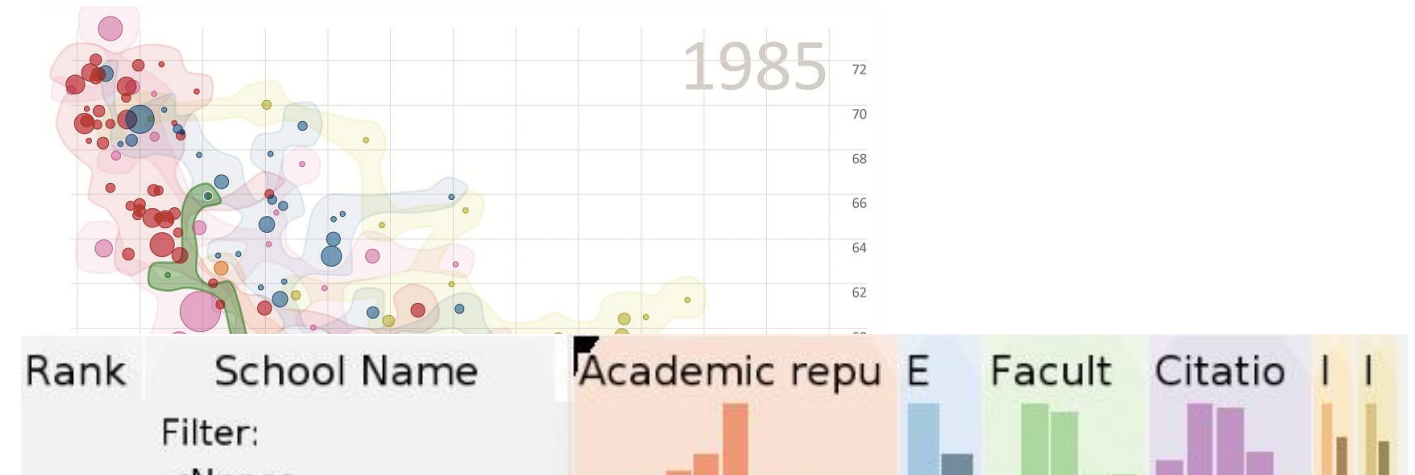
- Means to group multiple items together i.e. no explicit dependency
- **Sets** – **unordered** group of items
- **Lists** – group of items with **specified ordering**
- **Clusters** – grouping based on **attribute(s) similarity**
- Other complex structures:
 - Path – an ordered set of segments (in a network) formed by linking connecting nodes
 - Compound network – a network with associated tree, etc.
 - Hybrid networks



How we group items:

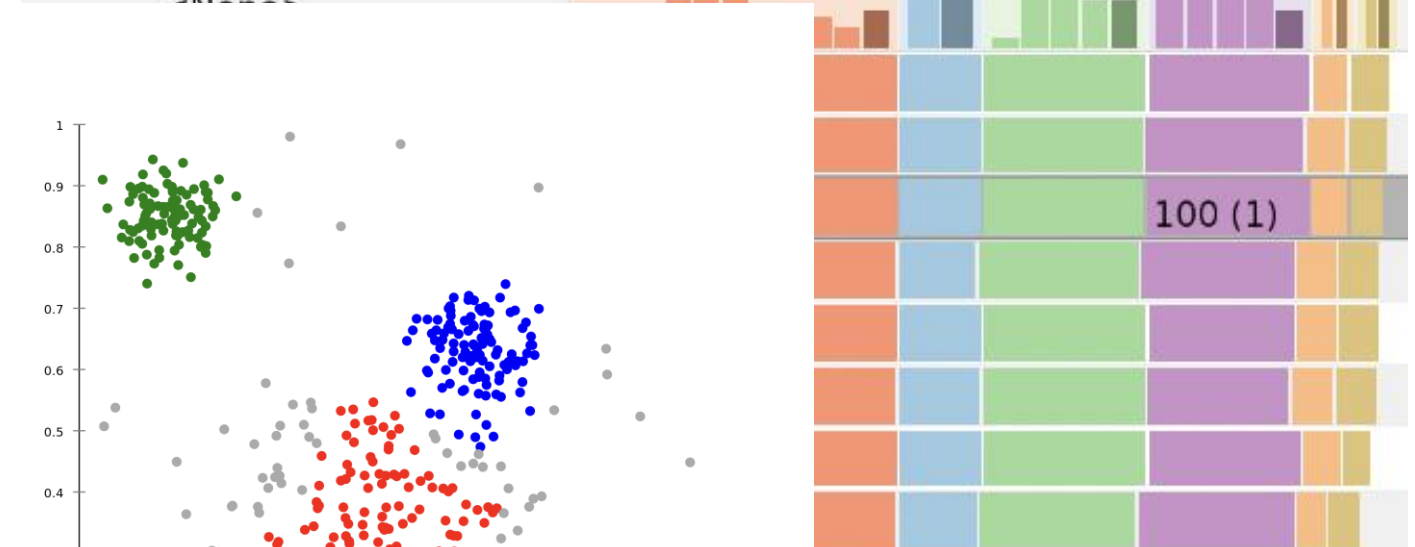
i. Sets

– unique items, unordered



ii. Lists

– ordered, duplicates possible

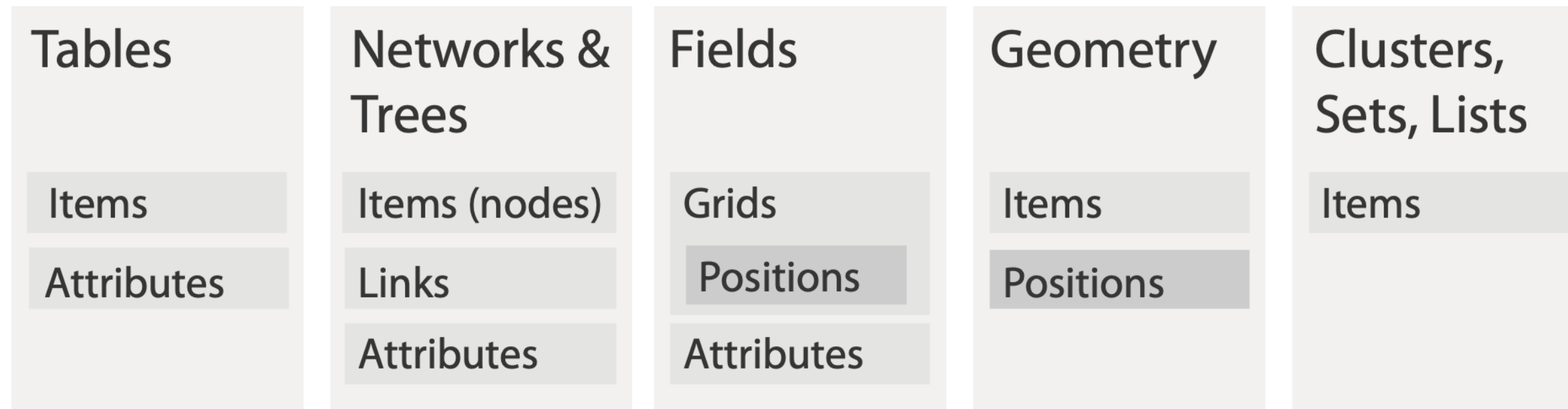


iii. Clusters

– groups of similar items



➔ Data and Dataset Types



➔ Data Types

➔ Items ➔ Attributes ➔ Links ➔ Positions ➔ Grids

- Which classes of values & measurements?
- **Categorical** (nominal)
 - compare equality
 - no implicit ordering
- **Ordered**
 - *ordinal*
 - less/greater than defined
 - *quantitative*
 - meaningful magnitude
 - arithmetic possible

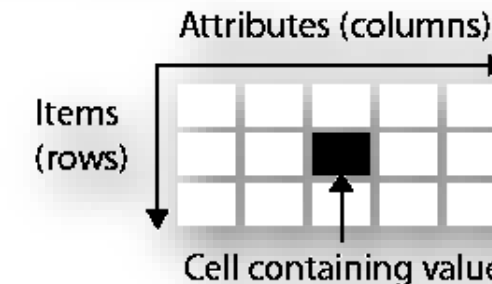


Class Activity

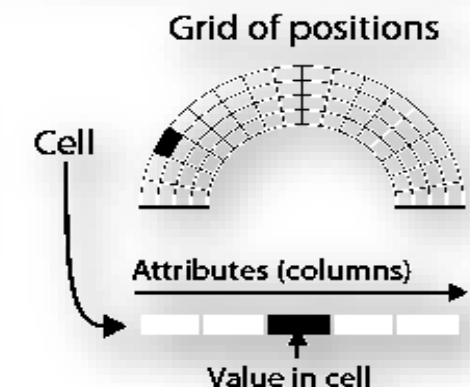
- This is about ‘meaning’ associated to attribute i.e. what its mean
- Focus only on the following:
 - Key vs Value
 - Continuous vs Discrete
 - Spatial vs Non-Spatial

- **Key** attribute is an *index* to look up **value** attribute; implicit or explicit (key value attribute)
- Row’s id (implicit) key to a flat table item; unique key value (unique) is an explicit key to an item
- Value attribute is the ‘entity’ contained in cell

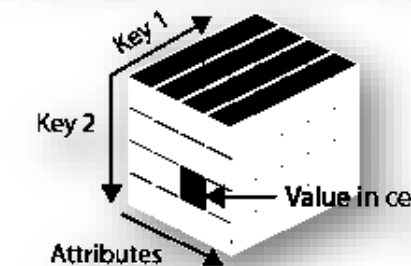
→ Tables



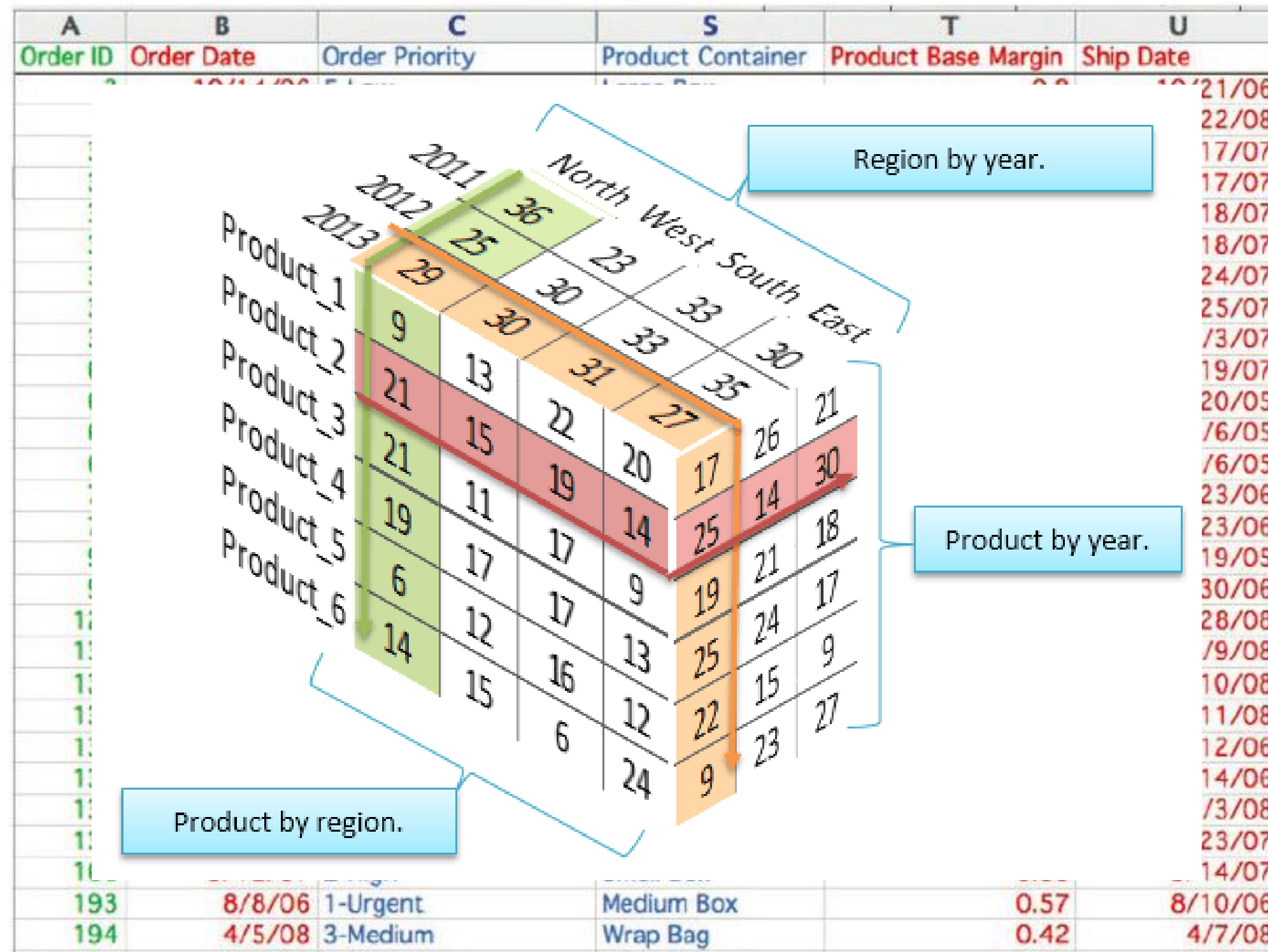
→ Fields (Continuous)



→ Multidimensional Table



Attributes Semantics of Table Dataset



Discrete (Non-Spatial) vs Continuous (Spatial)

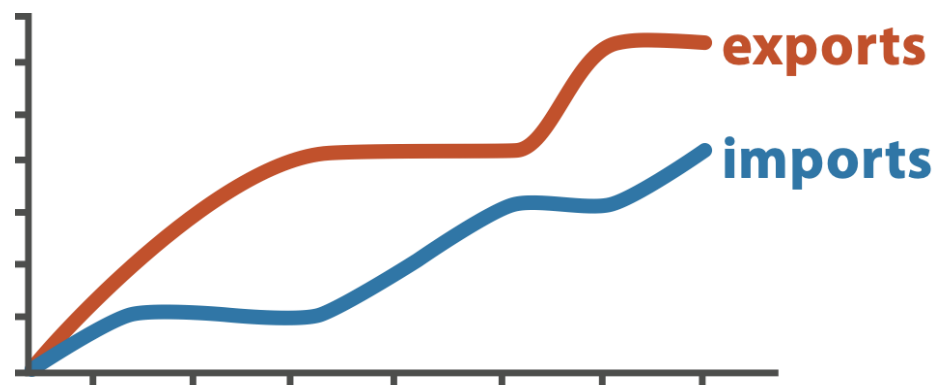
- Fields are used to describe dataset in continuous domains; hence fields' attribute value is quantitative.
- Accessing these values (in cells) though require indexing (key), typically using spatial positions (also quantitative).
- Also known as independent variable/dimension to acts as key and dependent variable is fields' attribute value.
- Example, temperature (dependent variable) recorded at a location A (spatial position in (x,y,z)) on earth's surface.
- Fields characterized in terms of number of key (dimension) vs values – multidimensional (multiple keys) like in 2D, 3D or 4D (time dimension); and multivariate like scalar, vector, tensor, etc.

- **Translate** from domain-specific language to generic visualization language
 - **Identify dataset type(s), attribute types**
 - **Identify cardinality**
 - how many items in the dataset?
 - what is cardinality of each attribute?
 - number of levels for categorical data
 - range for quantitative data
- *** *Consider whether to transform data*
- *guided by understanding of task*

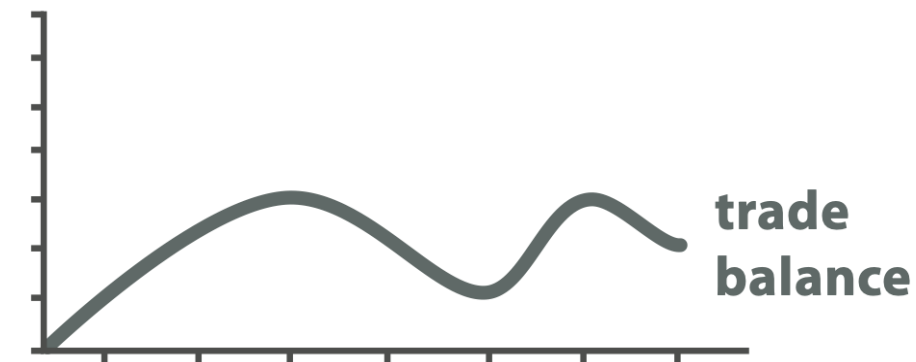
- **Data model**
 - mathematical abstraction
 - sets with operations, e.g., floats with $*$ / - +
 - variable data types in programming languages
- **Conceptual model**
 - mental construction (semantics)
 - supports reasoning
 - typically based on understanding of tasks [stay tuned!]
- **Data abstraction process relies on conceptual model**
 - for transforming data if needed

- **Data model: floats**
 - 32.52, 54.06, -14.35, ...
- **Conceptual model**
 - Temperature
- **Multiple possible data abstractions**
 - continuous to 2 significant figures: quantitative
 - *task: forecasting the weather*
 - hot, warm, cold: ordinal
 - *task: deciding if bath water is ready*
 - above freezing, below freezing: categorical
 - *task: decide if I should leave the house today*

- **Derived attribute:** compute from originals
 - simple change of type
 - acquire additional data
 - complex transformation



Original Data



$$\text{trade balance} = \text{exports} - \text{imports}$$

Derived Data

Analysis example: Derive One Attribute

- Strahler number
 - centrality metric for trees/networks
 - derived quantitative attribute
 - draw top 5K of 500K for good skeleton

[Using Strahler numbers for real time visual exploration of huge graphs. Auber.
Proc. Intl. Conf. Computer Vision and Graphics, pp. 56–69, 2002.]

